

RAM, ROM & Cache — Detailed Lecture Notes

JKSSB Computer Awareness

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08.1 Memory, volatility, and why the distinction matters

Main memory (also called primary memory) holds the data and instructions that the CPU uses directly. Secondary storage such as a hard disk or an SSD is not accessed by the CPU directly; its contents are brought into main memory first. This lesson concentrates on the primary-memory components a computer-awareness paper actually asks about: RAM, ROM, and cache.

The single most tested distinction in this lesson is **volatility**.

Definition (Volatile memory). *Memory whose stored contents are lost when the power supply is removed.*

Definition (Non-volatile memory). *Memory whose stored contents remain available after the power supply is removed.*

RAM is volatile; ROM is non-volatile. A practice item may phrase this as a NOT/incorrect-statement question, so learn both directions: “RAM loses its contents on power-off,” and “ROM retains its contents without power.”

08.2 RAM: volatile main memory

RAM stands for **Random Access Memory**. It is the main memory in which running programs and their active data are held. The term “random access” means that any storage location can be reached in roughly the same amount of time; it is not a statement about volatility. RAM is **volatile**: when the office desktop (EX-01) is switched off, whatever was in RAM is lost unless it was saved to secondary storage.

Example 08.2.1. *A stem states that RAM keeps its data intact after the computer is switched off. This is false. RAM is volatile, so its contents are lost on power-off; it is ROM (and flash) that retains data without power. If the stem instead asks which memory holds the running programs, select RAM.*

08.3 DRAM versus SRAM

RAM comes in two main types, and the exam routinely contrasts them. Both are RAM and both are volatile; the difference lies in speed, refresh, cost, and use.

Feature	DRAM	SRAM
Full form	Dynamic Random Access Memory	Static Random Access Memory
Refresh	Needs periodic refresh	No refresh needed
Speed	Slower	Faster
Density and cost	Denser and cheaper per bit	Less dense and costlier
Typical use	Main memory (the computer's RAM)	Cache memory

DRAM stores each bit in a tiny capacitor that leaks charge, so it must be **refreshed** periodically. That refresh overhead is the price of its high density and low cost, which is why DRAM is the natural choice for large main memory. **SRAM** uses a different cell design that holds its state without refresh, making it faster but more expensive and less dense; it is therefore used for the small, fast **cache**.

Example 08.3.1. *An item asks which RAM type is used to build cache memory. SRAM is faster and needs no refresh, so it is used for cache; DRAM is denser and cheaper and serves as main memory. If the stem asks which type needs periodic refreshing, the answer is DRAM.*

08.4 ROM and the ROM family

ROM stands for **Read-Only Memory**. It is **non-volatile** and is **read-only in normal operation**: the processor reads its contents, and changing them requires a special programming process rather than ordinary memory writes. ROM commonly stores **firmware**, such as the start-up instructions a computer runs before an operating system loads.

Definition (ROM). *Read-Only Memory: non-volatile primary memory that is normally read but not written during ordinary operation.*

Several ROM variants appear in exams. They differ mainly in *when* they are programmed and *how* they are erased.

Type	Key property
PROM	Programmable Read-Only Memory: can be programmed once by the user; it cannot be erased and reprogrammed.
EPROM	Erasable Programmable Read-Only Memory: erased by exposing the chip to ultraviolet (UV) light.
EEPROM	Electrically Erasable Programmable Read-Only Memory: erased and rewritten electrically , without removing the chip.
Flash	An EEPROM-derived non-volatile memory erased in blocks ; widely used in USB drives and SSDs.

Example 08.4.1. *An item asks which memory can be erased by ultraviolet light. PROM is one-time programmable and cannot be erased; EEPROM is erased electrically; flash is erased in blocks. The UV-erasable type is EPROM. If the stem instead asks which type is programmable only once, select PROM.*

08.5 Cache memory and its levels

Cache is a small, very fast memory placed between the CPU and main memory. It holds frequently or recently used data and instructions so the CPU does not have to wait as often for slower main memory. Cache is **faster than main memory** — a fact the paper has tested directly (PYQ-03). It is temporary working memory, **not** a backup and not a substitute for storage.

Cache is organised in levels.

Definition (L1 cache). *The level-1 cache: the smallest, fastest cache, located closest to the CPU core.*

Definition (L2 cache). *The level-2 cache: larger and slightly slower than L1.*

Definition (L3 cache). *The level-3 cache: the largest and slowest of the three levels, typically shared across the cores of the processor.*

As the level number rises, size grows and speed falls, and distance from the core increases: $L1 < L2 < L3$ in size, with the speed order reversed. Even L3, however, is faster than main memory.

08.6 Cache hits, misses, and hit ratio

Performance depends on how often the CPU finds what it needs already in the cache.

Definition (Cache hit). *A memory access in which the required data is found in the cache.*

Definition (Cache miss). *A memory access in which the required data is not found in the cache, so it must be fetched from main memory.*

The **hit ratio** measures how often accesses succeed in the cache:

$$\text{hit ratio} = \frac{\text{hits}}{\text{total accesses}}.$$

Example 08.6.1. *If the cache receives 100 memory accesses and 90 of them are found in the cache, then the hit ratio is $90/100 = 0.90$. The remaining 10 accesses are misses and must go to main memory. A higher hit ratio means the CPU waits less, so the system performs better.*

08.7 Exam retrieval and common confusions

Use the following distinctions to answer both recall and NOT/incorrect items reliably:

- RAM is **volatile**; ROM is **non-volatile** (TRAP-01).
- ROM is **read-only in normal operation**; writing needs a special process (TRAP-02).
- **Cache is faster** than main memory, not slower (PYQ-03).
- **SRAM** is used for cache; **DRAM** is used for main memory — and DRAM is the type that needs refresh.
- Cache is **not** a backup; it is temporary working memory.
- ROM family: PROM (once), EPROM (UV), EEPROM (electrical), flash (block erase).
- Hit ratio = hits $\div$ total accesses.

Exercises

1. Explain why RAM is described as volatile and ROM as non-volatile.
2. Distinguish DRAM from SRAM in terms of speed, refresh, and typical use.
3. Expand PROM, EPROM, EEPROM, and flash, and state how each is erased (or whether it can be).
4. Order L1, L2, and L3 cache by size and by speed, and state which level is typically shared across cores.
5. A cache handles 200 memory accesses and 150 are hits. Compute the hit ratio.

Solutions

1. RAM is volatile because it loses its contents when power is removed; ROM is non-volatile because it retains its contents without power.
2. DRAM needs periodic refresh, is slower but denser and cheaper, and is used for main memory. SRAM needs no refresh, is faster but costlier and less dense, and is used for cache.
3. PROM: Programmable Read-Only Memory, programmable once and not erasable. EPROM: Erasable Programmable Read-Only Memory, erased by ultraviolet light. EEPROM: Electrically Erasable Programmable Read-Only Memory, erased electrically. Flash: EEPROM-derived, erased in blocks.
4. By size: $L1 < L2 < L3$, so L3 is the largest. By speed: L1 is fastest and L3 the slowest of the three. L3 is typically shared across cores.
5. Hit ratio = $150/200 = 0.75$.